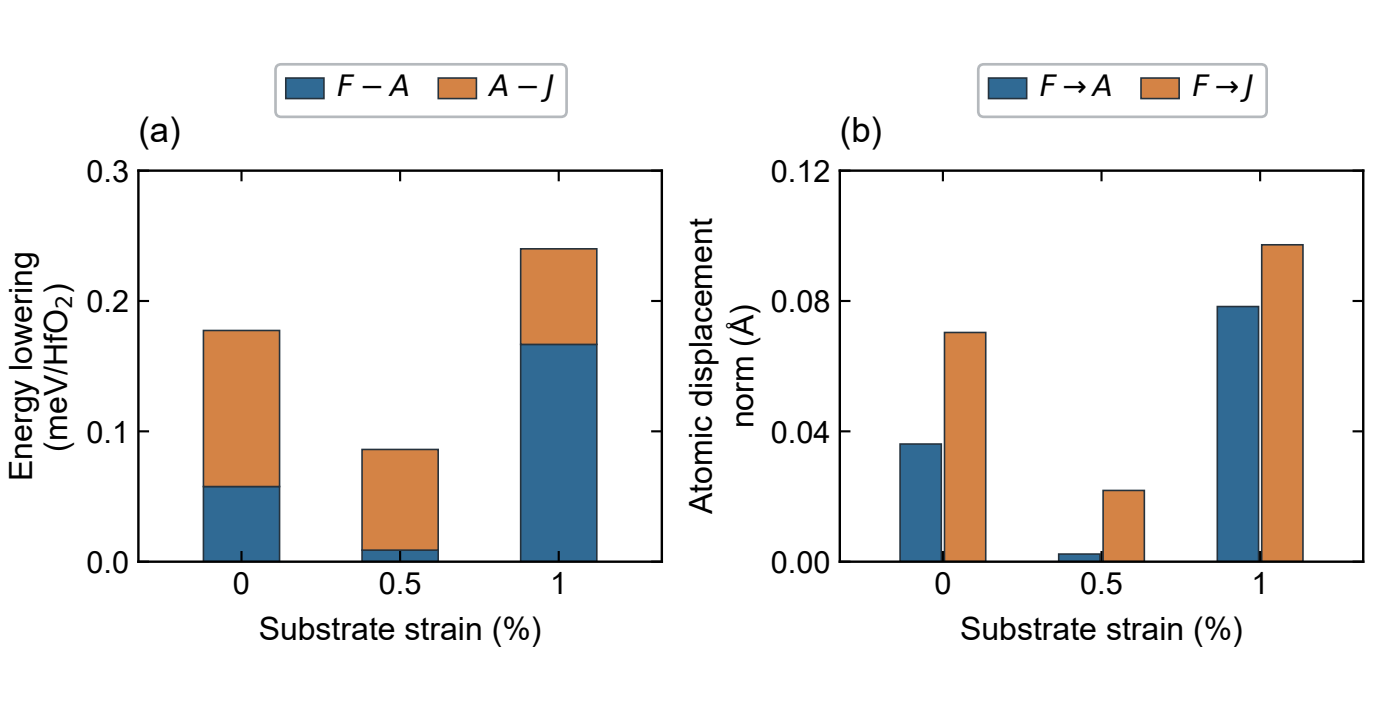
(a)
F − A
A − J
Energy lowering (meV/HfO2)
0.3
0.2
0.1
0.0
0
0.5
1
Substrate strain (%)
(b)
F → A
F → J
Atomic displacement norm (Å)
0.12
0.08
0.04
0.00
0
0.5
1
Substrate strain (%)